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Film removal apparatus and uncovering mechanism

Abstract

This disclosure relates to a film removal apparatus that is suitable for uncovering a film of a semiconductor substrate and includes a carrier, an uncovering mechanism and a lifting mechanism. The carrier supports the semiconductor substrate. The uncovering mechanism includes a slider, a guider and a roller. The slider is slidably disposed on the carrier. The guider is fixed to the slider. The roller is rotatably disposed on the slider and arranged beside the guider. The uncovering mechanism is to uncover the film from the semiconductor substrate. The lifting mechanism selectively separates the semiconductor substrate away from the carrier.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This non-provisional application claims priority under 35 U.S.C. § 119 (a) on Patent

TECHNICAL FIELD

(2) The present disclosure relates to a film removal apparatus and an uncovering mechanism.

BACKGROUND

(3) In general, a semiconductor substrate is placed on a frame so as to be easily transferred. Since the semiconductor substrate is typically supported by the frame through attachment of a film, the film needs to be peeled off the frame and the semiconductor substrate for detaching the semiconductor substrate from the frame. However, the film is conventionally peeled off in manual, which is time-consuming and may easily damage the component on the semiconductor substrate due to uneven peeling force.

SUMMARY

(4) The present disclosure provides a film removal apparatus and an uncovering mechanism capable of efficiently, easily and evenly peeling a film off a semiconductor substrate while preventing damage of the semiconductor substrate or any component thereon.

(5) According to one aspect of the present disclosure, a film removal apparatus suitable for uncovering a film of a semiconductor substrate includes a carrier, an uncovering mechanism and a lifting mechanism. The carrier supports the semiconductor substrate. The uncovering mechanism includes a slider, a guider and a roller. The slider is slidably disposed on the carrier. The guider is fixed to the slider. The roller is rotatably disposed on the slider and arranged beside the guider. The uncovering mechanism is to uncover the film from the semiconductor substrate. The lifting mechanism selectively separates the semiconductor substrate away from the carrier.

(6) According to the film removal apparatus and the uncovering mechanism discussed above, the uncovering mechanism provided with the slider, the guider and the roller can uncover the film of the semiconductor substrate with a stable force, and mechanical operation thereof instead of manual work prevents accidental injury of human.

(7) Further, the semiconductor substrate is lifted by the lifting mechanism, which prevents damage of the semiconductor substrate due to improper external force applied thereon during the lifting process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only and thus are not intending to limit the present disclosure and wherein:

(2) FIG. 1 is a perspective view of a film removal apparatus according to one embodiment of the present disclosure;

(3) FIG. 2 is an exploded view of a base of a film removal apparatus according to one embodiment of the present disclosure and a semiconductor substrate;

(4) FIG. 3 is a partial and perspective view of a carrier of a film removal apparatus according to one embodiment of the present disclosure;

(5) FIG. 4 is a perspective view of an uncovering mechanism of a film removal apparatus according to one embodiment of the present disclosure;

(6) FIG. 5 is a perspective view of a lifting mechanism of a film removal apparatus according to one embodiment of the present disclosure;

(7) FIG. 6 is a perspective view of a carrier a film removal apparatus according to one embodiment of the present disclosure with the base been omitted;

(8) FIG. 7 to FIG. 13 are schematic views showing the action of a film removal apparatus

according to one embodiment of the present disclosure;

(9) FIG. **14** is a perspective view of racks of a carrier and an uncovering mechanism according to another embodiment of the present disclosure;

(10) FIG. **15** to FIG. **16** are schematic views showing the action of an uncovering mechanism according to another embodiment of the present disclosure;

(11) FIG. **17** is a perspective view of a carrier and a lifting mechanism according to further another embodiment of the present disclosure;

(12) FIG. **18** is a perspective view of a carrier and a lifting mechanism according to further another embodiment of the present disclosure with a pusher been omitted;

(13) FIG. **19** is a perspective view showing the rotation of an output gear with respect to racks of a lifting mechanism according to further another embodiment of the present disclosure; and

(14) FIG. **20** and FIG. **21** are schematic views showing the action of a lifting mechanism according to further another embodiment of the present disclosure.

DETAILED DESCRIPTION

(15) Aspects and advantages of the invention will become apparent from the following detailed descriptions with the accompanying drawings. For purposes of explanation, one or more specific embodiments are given to provide a thorough understanding of the invention, and which are described in sufficient detail to enable one skilled in the art to practice the described embodiments. It should be understood that the following descriptions are not intended to limit the embodiments to one specific embodiment. On the contrary, it is intended to cover alternatives, modifications, and equivalents as can be included within the spirit and scope of the described embodiments as defined by the appended claims.

(16) Please refer to FIG. **1** and FIG. **2**, where FIG. **1** is a perspective view of a film removal apparatus **10** according to one embodiment of the present disclosure, and FIG. **2** is an exploded view of a base **110** of the film removal apparatus **10** in FIG. **1** and a semiconductor substrate **20**.

(17) The film removal apparatus **10** provided in this embodiment is suitable for removing a film **24** of the semiconductor substrate **20**. For example, the semiconductor substrate **20** may include a holder **21**, a plurality of electronic parts **22**, a fixed frame **23** and the film **24**. The electronic parts **22** are stacked on the holder **21**. The fixed frame **23** surrounds the holder **21**, and the holder **21** holding the electronic parts **22** is fixed with respect to the fixed frame **23** through adhesion of the film **24**.

(18) The film removal apparatus **10** includes a carrier **100**, an uncovering mechanism **200** and a lifting mechanism **300**. The carrier **100** supports the semiconductor substrate **20** and includes the base **110**, two lateral movement components **120** and a plurality of holding nozzles **130**. The base **110** may, for example, match the semiconductor substrate **20** in shape so as to support the semiconductor substrate **20**. The base **110** has a plurality of holes **111** and two notches **112**. The lateral movement components **120** are slidably disposed on the base **110** between a closed position (as shown in FIG. **12**) and an open position (as shown in FIG. **13**). As shown in FIG. **12**, when the lateral movement components **120** are located at the closed position, the lateral movement components **120** are respectively located in the notches **112**. As shown in FIG. **13**, when the lateral movement components **120** are located at the open position, the lateral movement components **120** are at least partially located outside the notches **112**, respectively. The holding nozzles **130** are disposed on the lateral movement components **120**, and negative pressure may be generated at the holding nozzles **130** and the holes **111** to suck and fix the semiconductor substrate **20**.

(19) Please be noted that the quantities of the notches **112** and the lateral movement components **120** located at two opposite sides of the base **110** in this embodiment are not intended to restrict the present disclosure. In some other embodiments of the present disclosure, there may be only one notch and one lateral movement component, and both of them are located at one side of the base.

(20) Please further refer to FIG. **3**, which is a partial and perspective view of the carrier **100** of the film removal apparatus **10** in FIG. **1**. The lateral movement components **120** are disposed in the

carrier **100** via, for example, pneumatic cylinders **150**. It can be also considered that the lateral movement components **120** are driven by the pneumatic cylinders **150** to be moved with respect to the base **110**.

(21) Please further refer to FIG. **4**, which is a perspective view of the uncovering mechanism **200** of the film removal apparatus **10** in FIG. **1**. The uncovering mechanism **200** includes two sliders **210**, a guider **220** and a roller **230**. The sliders **210** are slidably disposed on the carrier **100**. The guider **220** is fixed to the sliders **210**. The roller **230** is rotatably disposed on the sliders **210** and is arranged beside the guider **220**. In order to smoothen the sliding of the sliders **210** with respect to the carrier **100**, the uncovering mechanism **200** may, for example but not limited thereto, further include a plurality of wheels **240** rotatably disposed on the sliders **210** and placed on the carrier **100**. Moreover, in order to smoothen the rotation of the roller **230** with respect to the sliders **210**, the roller **230** may be, for example but not limited thereto, disposed on the sliders **210** via bearings (not shown).

(22) Please further refer to FIG. **5**, which is a perspective view of the lifting mechanism **300** of the film removal apparatus **10** in FIG. **1**. The lifting mechanism **300** may include a movement assembly **310** and a plurality of transfer nozzles **320**. The movement assembly **310** includes a fixed stand **311**, a movable stand **312**, a driving motor **313**, a transmission set **314** and a nozzle holder **315**. The movable stand **312** is movably disposed with respect to the fixed stand **311**. The driving motor **313** is disposed on the fixed stand **311** and is connected to the movable stand **312** via the transmission set **314**. The nozzle holder **315** may be connected to the movable stand **312** via, for example, elastic elements (e.g., springs). As such, during the operation of the driving motor **313**, the transmission set **314** can be driven to move the movable stand **312** with respect to the fixed stand **311**. In this embodiment, the transmission set **314** may be, for example but not limited thereto, a screw and a nut that match each other, such that the rotational motion of the driving motor **313** can be converted into linear motion of the movable stand **312** through the screw and the nut. The transfer nozzles **320** are disposed on the nozzle holder **315** to be lifted or lowered along with the nozzle holder **315**. The transfer nozzles **320** selectively sucks and fixes the semiconductor substrate **20** and thus separates the semiconductor substrate **20** away from the base **110** of the carrier **100**.

(23) Please be noted that the movement manner of the movable stand **312** and the nozzle holder **315** with respect to the fixed stand **311** through the driving motor **313** and the transmission set **314** in this embodiment is not intended to restrict the present disclosure. In some other embodiments, the lifting mechanism may be wholly lifted or lowered with respect to the carrier of the film removal apparatus in a reasonable and proper manner.

(24) In this embodiment, besides the movement assembly **310** and the transfer nozzles **320** set over the carrier **100**, the lifting mechanism **300** may separate the semiconductor substrate **20** away from the base **110** of the carrier **100** through a different lifting manner. Please further refer to FIG. **6**, which is a perspective view of the carrier **100** in FIG. **3** with the base **110** been omitted for convenience of observation. The lifting mechanism **300** further includes a pusher **340** movably disposed in the carrier **100**. As such, when the pusher **340** is lifted with respect to the base **110**, the pusher **340** is able to push up the semiconductor substrate **20** to be away from the carrier **100**. Moreover, the pusher **340** in this embodiment is disposed in the carrier **100** through, for example, pneumatic cylinders **350**. It can be also considered that the pusher **340** is driven by the pneumatic cylinders **350** to be moved with respect to the base **110**.

(25) Please be noted that the lifting manners achieved by the cooperation of the movement assembly **310** and the transfer nozzles **320** located over the carrier **100** and the cooperation of the pusher **340** and the pneumatic cylinders **350** disposed in the carrier **100** in this embodiment are not intended to restrict the present disclosure. In some embodiments of the present disclosure, the semiconductor substrate may be separated away from the carrier in just one lifting manner, such as the movement assembly and the transfer nozzles over the carrier, the pusher and the pneumatic

cylinders in the carrier, or another reasonable and proper manner.

(26) Please refer to FIG. 7 to FIG. 13, which are schematic views showing the action of the film removal apparatus **10** in FIG. 1. First, as shown in FIG. 7, an adhesive tape TP has one end adhered to the film **24** of the semiconductor substrate **20** and another end adhered to the roller **230** via the guider **220** of the uncovering mechanism **200** before the uncovering process. That is, the film **24** of the semiconductor substrate **20** is connected to the roller **230** of the uncovering mechanism **200** via the adhesive tape TP. Please be noted that the adhesive tape TP can be adhered manually or mechanically, and the present disclosure is not limited thereto.

(27) Then, as shown in FIG. 8, the uncovering mechanism **200** is pushed along a direction AA. During the movement of the uncovering mechanism **200**, the film **24** of the semiconductor substrate **20** is forced to be gradually separated from the holder **21** and the fixed frame **23** of the semiconductor substrate **20**, and the film **24** separated from the holder **21** and the fixed frame **23** of the semiconductor substrate **20** can be wound on the roller **230** through rotation of the roller **230** along a direction BB. By doing so, the uncovering mechanism **200** of this embodiment can uncover the film **24** of the semiconductor substrate **20** with a stable force, and mechanical operation thereof instead of manual work prevents accidental injury of human.

(28) Then, as shown in FIG. 9 and FIG. 10, after the film **24** is completely peeled off, the transfer nozzles **320** of the lifting mechanism **300**, for example, can be used to suck and fix the holder **21** and the fixed frame **23** of the semiconductor substrate **20** (along a direction CC), such that the holder **21** holding the electronic parts **22** and the fixed frame **23** can be lifted away from the base **110**, and the rest part (not shown) of the semiconductor substrate **20** is remained on the base **110**. Then, the fixed frame **23** and the holder **21** holding the electronic parts **22** can be sequentially removed from the transfer nozzles **320**. Please be noted that in some other embodiments of the present disclosure, the transfer nozzles of the lifting mechanism can only suck and fix the holder of the semiconductor substrate, while the fixed frame and the rest part of the semiconductor substrate are remained on the base.

(29) Please be noted that there is also another lifting manner achieved by the cooperation of the pusher **340** and the pneumatic cylinders **350** disposed in the carrier **100** in this embodiment. Therefore, the pusher **340** and the pneumatic cylinders **350** disposed in the carrier **100** can also be selected to separate the holder **21** holding the electronic parts **22** of the semiconductor substrate **20** away from the base **110**.

(30) In detail, as shown in FIG. 11, the film **24** of the semiconductor substrate **20** is already removed from the holder **21** and the fixed frame **23** of the semiconductor substrate **20**. Then, as shown in FIG. 12, the fixed frame **23** can be removed along a direction DD, and then the pusher **340** can be lifted along a direction EE through the driving of the pneumatic cylinders **350**. As such, the pusher **340** is able to push up the holder **21** holding the electronic parts **22** of the semiconductor substrate **20** so as to separate the holder **21** holding the electronic parts **22** from the base **110**.

(31) Then, as shown in FIG. 13, the lateral movement components **120** are moved outside the notches **112** of the base **110** along directions FF by the driving of the pneumatic cylinders **150**, so that the spaces in the notches **112** can be exposed for convenience of the careful removal of the holder **21** holding the electronic parts **22** (along a direction GG). Accordingly, the problems existing in convention of which the semiconductor substrate is difficult to be directly removed from the base and the semiconductor substrate is easily damaged due to uneven force applied thereon can be solved.

(32) Please refer to FIG. 14, which is a perspective view of racks **160** of a carrier **100A** and an uncovering mechanism **200A** according to another embodiment of the present disclosure. Note that this embodiment is similar to the previous embodiment, and thus differences between them and necessary elements will be mainly illustrated hereinafter. In this embodiment, the carrier **100A** further includes two racks **160**. The uncovering mechanism **200A** includes two sliders **210**, a guider **220**, a roller **230**, a plurality of wheels **240**, a rolling contactor **250** and two gear trains **260**. The

sliders **210** are slidably disposed on the carrier **100A**. The guider **220** is fixed to the sliders **210**. The roller **230** is rotatably disposed on the sliders **210** and is arranged beside the guider **220**. The wheels **240** are rotatably disposed on the slider **210** and placed on the carrier **100A**. The rolling contactor **250** is rotatably disposed on the sliders **210**, and the guider **220** is located between the roller **230** and the rolling contactor **250**. The gear trains **260** are rotatably disposed on the sliders **210**, respectively. Each gear train **260** includes a first transmission gear **261** and a second transmission gear **262**. The first transmission gears **261** are respectively meshed with the racks **160**. The second transmission gears **262** may be, for example, coaxially fixed to the roller **230**, and are respectively meshed with the first transmission gears **261**. Accordingly, when the sliders **210** moves with respect to the racks **160**, the first transmission gears **261** are rotated with respect to the sliders **210** through the action of the racks **160**, and the second transmission gears **262** are driven by the first transmission gears **261** to rotate the roller **230** with respect to the sliders **210**.

(33) In this embodiment and some other embodiments of the present disclosure, the uncovering mechanism **200A** may further include two pushed pieces **270** respectively fixed on the sliders **210**. As such, the pushed pieces **270** can be pushed by a mechanical arm **30** (shown in FIG. 15) to achieve the mechanical uncovering process.

(34) Please be noted that the quantities of the racks **160** and the gear trains **260** in this embodiment are not intended to restrict the present disclosure. In some other embodiments of the present disclosure, there may be only one rack and one gear train.

(35) Please be noted that the pushed pieces **270** in this embodiment are optional, and the present disclosure is not limited thereto. In some other embodiments of the present disclosure, there may be only one pushed piece or no pushed piece, and the uncovering mechanism can be pushed by the mechanical arm via the sliders or the rolling contactor, for example.

(36) Please further refer to FIG. 15 and FIG. 16, which are schematic views showing the action of the uncovering mechanism **200A** in FIG. 14. As shown in FIG. 15, an adhesive tape TP has one end adhered to the film **24** of the semiconductor substrate **20** and another end adhered to the roller **230** via the guider **220** of the uncovering mechanism **200A** before the uncovering process. That is, the film **24** of the semiconductor substrate **20** is connected to the roller **230** of the uncovering mechanism **200A** via the adhesive tape TP.

(37) Then, as shown in FIG. 16, the pushed pieces **270** of the uncovering mechanism **200A** is pushed by the mechanical arm **30** along a direction HH. During the movement of the uncovering mechanism **200A**, the film **24** of the semiconductor substrate **20** is forced to be gradually separated from the holder **21** and the fixed frame **23** of the semiconductor substrate **20**, the roller **230** is rotated along a direction II by the transmission force provided through the racks **160**, the first transmission gears **261** and the second transmission gears **262**, and the film **24** separated from the holder **21** and the fixed frame **23** of the semiconductor substrate **20** can be wound on the roller **230** through rotation of the roller **230**. By doing so, the uncovering mechanism **200A** of this embodiment can uncover the film **24** of the semiconductor substrate **20** with a stable force, and mechanical operation thereof instead of manual work prevents accidental injury of human.

Moreover, the rolling contactor **250** can, for example, gently press against the holder **21** and the fixed frame **23** where the film **24** is already removed during the uncovering process, so that the holder **21** and the fixed frame **23** can be prevented from being unwantedly pulled up by the film **24**.

(38) Please refer to FIG. 17 to FIG. 19, where FIG. 17 is a perspective view of a carrier **100** and a lifting mechanism **300A** according to further another embodiment of the present disclosure, FIG. 18 is a perspective view of the carrier **100** and the lifting mechanism **300A** in FIG. 17 with a pusher **340** been omitted for convenience of observation, and FIG. 19 is a perspective view showing the rotation of an output gear **376** with respect to racks **371** of the lifting mechanism **300A** in FIG. 17. Note that this embodiment is similar to the previous embodiment, and thus differences between them and necessary elements will be mainly illustrated hereinafter.

(39) The lifting mechanism **300A** includes the pusher **340** movably disposed in the carrier **100**

(numbered in FIG. 6). The holding nozzles **130** are fixed to the lateral movement components **120A** and **120B**. Moreover, the lifting mechanism **300A** further includes two racks **371** and **372**, two abutting components **373** and **374** and a driving motor **375**. The racks **371** and **372** are disposed in the carrier **100**. The lateral movement component **120A** and the abutting component **373** are disposed on the rack **371**, such that the lateral movement component **120A**, the abutting component **373** and the rack **371** can be moved together. Also, the lateral movement component **120B** and the abutting component **374** are disposed on the rack **372**, such that the lateral movement component **120B**, the abutting component **374** and the rack **372** can be moved together. The driving motor **375** is disposed on the carrier **100** and has an output gear **376** meshed with the racks **371** and **372**. As shown in FIG. 19, when the output gear **376** rotates along a direction JJ, the racks **371** and **372** are driven by the output gear **376** to be moved respectively along directions KK and LL, and then the lateral movement components **120A** and **120B** and the abutting components **373** and **374** are driven by the racks **371** and **372** to be moved with respect to the carrier **100**. Moreover, the abutting components **373** and **374** respectively have inclined surfaces **3731** and **3741**. The pusher **340** can be abutted by the abutting components **373** and **374** through the inclined surfaces **3731** and **3741** so as to be lifted or lowered with respect to the carrier **100**.

(40) Please be noted that the quantities of the racks **371** and **372** and the abutting components **373** and **374** are not intended to restrict the present disclosure. In some other embodiments of the present disclosure, there may be only one rack and one abutting component.

(41) Please refer to FIG. 20 and FIG. 21, which are schematic views showing the action of the lifting mechanism **300A** in FIG. 17. As shown in FIG. 20, the film **24** of the semiconductor substrate **20** is already removed from the holder **21** and the fixed frame **23** of the semiconductor substrate **20**. Then, as shown in FIG. 21, the fixed frame **23** can be removed along a direction MM, and then the driving motor **375** rotates the output gear **376** along a direction NN. As such, the output gear **376** drives the racks **371** and **372** to move along directions OO and PP, respectively. By doing so, the lateral movement component **120A** and the abutting component **373** are driven by the rack **371** to be moved along the direction OO, and the lateral movement component **120B** and the abutting component **374** are driven by the rack **372** to be moved along the direction PP.

Accordingly, the lateral movement components **120A** and **120B** are moved away from each other to be located outside the notches **112**, and the abutting components **373** and **374** are moved towards each other to lift the pusher **340** along a direction QQ through the inclined surfaces **3731** and **3741** of the abutting components **373** and **374**.

(42) Then, since the lateral movement components **120A** and **120B** are moved outside the notches **112** of the base **110**, the spaces in the notches **112** are exposed for convenience of the careful removal of the holder **21** holding the electronic parts **22**. Accordingly, the problems existing in convention of which the semiconductor substrate is difficult to be directly removed from the base and the semiconductor substrate is easily damaged due to uneven force applied thereon can be solved.

(43) According to the film removal apparatus and the uncovering mechanism discussed above, the uncovering mechanism provided with the slider, the guider and the roller can uncover the film of the semiconductor substrate with a stable force, and mechanical operation thereof instead of manual work prevents accidental injury of human.

(44) Further, the semiconductor substrate can be lifted by the cooperation of the movement assembly and the transfer nozzles located over the carrier, the cooperation of the pusher and the pneumatic cylinders disposed in the carrier, or the cooperation of the pusher, the racks and the abutting components disposed in the carrier, which prevents damage of the semiconductor substrate due to improper external force applied thereon during the lifting process.

(45) Furthermore, the movement of the lateral movement components outside the notches of the base is able to expose the spaces in the notches, which is convenient for the careful removal of the holder holding the electronic parts. Accordingly, the problems existing in convention of which the

semiconductor substrate is difficult to be directly removed from the base and the semiconductor substrate is easily damaged due to uneven force applied thereon can be solved.

(46) The embodiments are chosen and described in order to best explain the principles of the present disclosure and its practical applications, to thereby enable others skilled in the art best utilize the present disclosure and various embodiments with various modifications as are suited to the particular use being contemplated. It is intended that the scope of the present disclosure is defined by the following claims and their equivalents.

Claims

1. A film removal apparatus, suitable for uncovering a film of a semiconductor substrate, the film removal apparatus comprising: a carrier, supporting the semiconductor substrate; an uncovering mechanism, comprising a slider, a guider and a roller, wherein the slider is slidably disposed on the carrier, the guider is fixed to the slider, and the roller is rotatably disposed on the slider and arranged beside the guider, and the uncovering mechanism is to uncover the film from the semiconductor substrate; and a lifting mechanism, selectively separating the semiconductor substrate away from the carrier.
2. The film removal apparatus according to claim 1, wherein the carrier comprises a plurality of holding nozzles selectively to suck and fix the semiconductor substrate.
3. The film removal apparatus according to claim 2, wherein the carrier further comprises a base and a lateral movement component, the base comprises a notch, the lateral movement component is slidably disposed on the base between a closed position and an open position, and the plurality of holding nozzles are disposed on the lateral movement component, the lateral movement component is located in the notch when the lateral movement component is located at the closed position, and the lateral movement component is at least partially located outside the notch when the lateral movement component is located at the open position.
4. The film removal apparatus according to claim 3, wherein the carrier comprises two lateral movement components, the base comprises two notches, and the notches are respectively located at two opposite sides of the base.
5. The film removal apparatus according to claim 3, wherein the lifting mechanism comprises a pusher, and the pusher is movably disposed in the carrier to selectively separate the semiconductor substrate away from the carrier.
6. The film removal apparatus according to claim 5, wherein the lateral movement component and the pusher are driven by a pneumatic cylinder so as to be moved with respect to the base.
7. The film removal apparatus according to claim 5, wherein the lifting mechanism further comprises a rack, an abutting component and a driving motor, the rack is disposed on the carrier, the lateral movement component and the abutting component are disposed on the rack, the driving motor is disposed on the carrier and comprises an output gear, the output gear is meshed with the rack, the lateral movement component and the abutting component are driven by the driving motor through the output gear and the rack so as to be moved with respect to the carrier, the abutting component comprises an inclined surface, and the pusher is abutted by the abutting component through the inclined surface so as to be lifted or lowered with respect to the carrier.
8. The film removal apparatus according to claim 7, wherein the lifting mechanism comprises two racks and two abutting components, the carrier comprises two lateral movement components, the output gear is meshed with the racks, the lateral movement components are respectively disposed on the racks, and the abutting components are respectively disposed on the racks.
9. The film removal apparatus according to claim 3, wherein the lifting mechanism comprises a movement assembly and a plurality of transfer nozzles, and the plurality of transfer nozzles are disposed on the movement assembly to selectively suck and separate the semiconductor substrate away from the base of the carrier.

10. The film removal apparatus according to claim 1, wherein the uncovering mechanism further comprises a plurality of wheels, and the plurality of wheels are rotatably disposed on the slider and placed on the carrier.
 11. The film removal apparatus according to claim 1, wherein the carrier comprises a base and a rack, the rack is disposed on the base, the uncovering mechanism further comprises a gear train rotatably disposed on the slider, and the gear train comprises a first transmission gear meshed with the rack and a second transmission gear fixed to the roller and meshed with the first transmission gear.
 12. The film removal apparatus according to claim 11, wherein the second transmission gear and the roller are coaxial.
 13. The film removal apparatus according to claim 1, wherein the uncovering mechanism further comprises a rolling contactor rotatably disposed on the slider, and the guider is located between the roller and the rolling contactor.
 14. The film removal apparatus according to claim 1, wherein the uncovering mechanism further comprises a pushed piece fixed on the slider and selectively pushed by a mechanical arm.
 15. An uncovering mechanism, suitable for uncovering a film of a semiconductor substrate located on a carrier, the uncovering mechanism comprising: a slider, slidably disposed on the carrier; a guider, fixed to the slider; a roller, rotatably disposed on the slider and arranged beside the guider; and a plurality of wheels, rotatably disposed on the slider and placed on the carrier.
 16. The uncovering mechanism according to claim 15, further comprising a gear train rotatably disposed on the slider, wherein the gear train comprises a first transmission gear meshed with a rack of the carrier and a second transmission gear fixed to the roller and meshed with the first transmission gear.
 17. The uncovering mechanism according to claim 16, wherein the second transmission gear and the roller are coaxial.
 18. The uncovering mechanism according to claim 15, further comprising a rolling contactor rotatably disposed on the slider, wherein the guider is located between the roller and the rolling contactor.
 19. The uncovering mechanism according to claim 15, further comprising a pushed piece fixed on the slider and selectively pushed by a mechanical arm.
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